Snowflake Integration with ServiceNow Using REST API, OAuth 2.0 and the Scheduled Job

This document provides a comprehensive guide for implementing OAuth integration between Snowflake and ServiceNow. It covers the complete process from user creation to testing, including detailed steps for OAuth token generation, REST API setup, and the implementation of a scheduled job to update Windows Server CI records in the CMDB.
For a detailed video, please check out our YouTube channel: https://www.youtube.com/watch?v=si-yGGUKrAo. Please like, share, and subscribe to our channel and Snowflake Integration with ServiceNow Using REST A... - ServiceNow Community

Integration Process / Steps:

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User Creation

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02

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Configuring security settings for OAuth

03

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Preparing REST API messages for communication

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Configuring OAuth settings within ServiceNow

05

REST Message Configuration

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06

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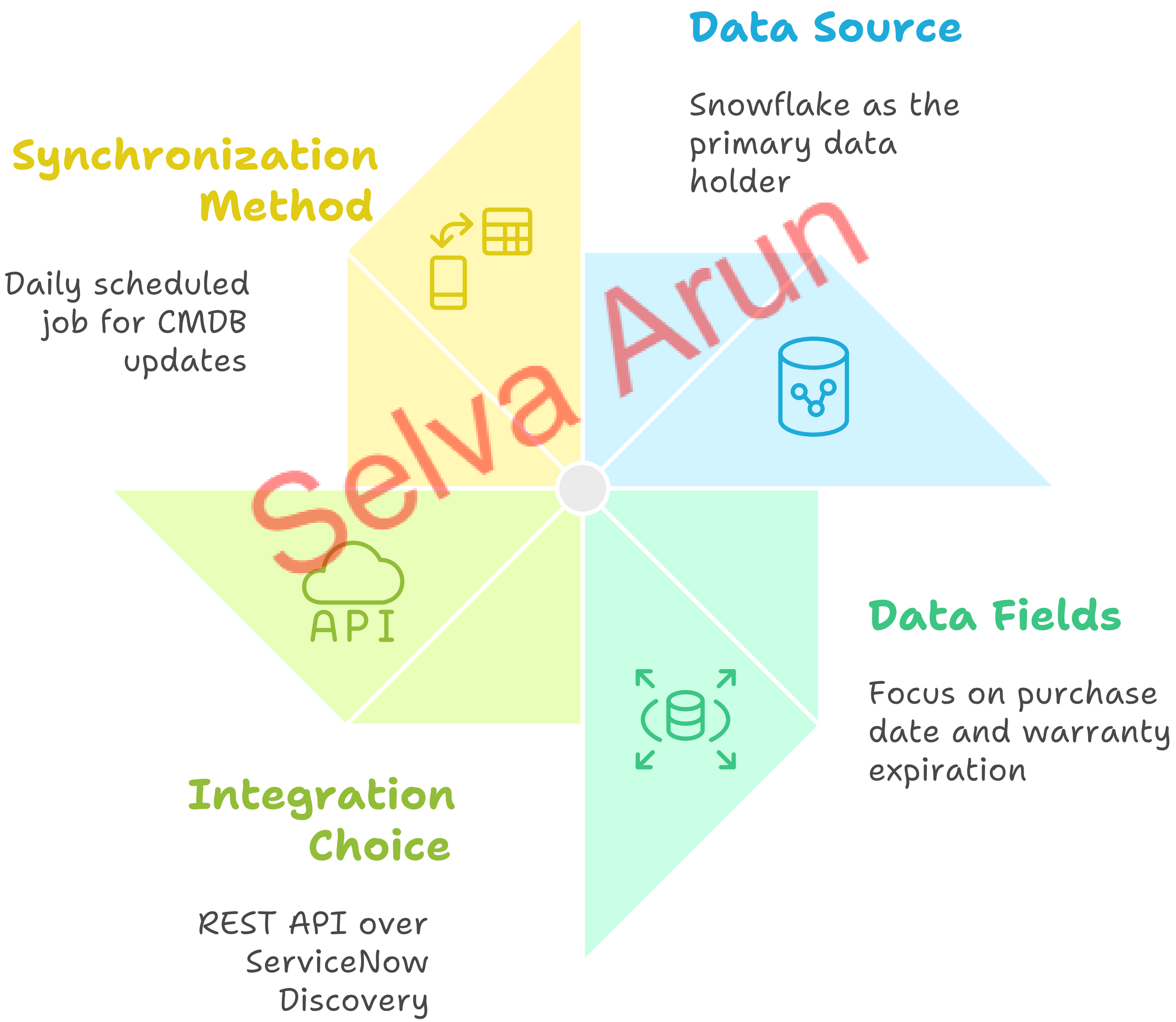
Use Case

The purpose of this integration is to enable Snowflake to ServiceNow integration using REST API instead of JDBC. The integration leverages OAuth 2.0 for secure authentication and ensures that the Windows Server CI records in the ServiceNow CMDB are regularly updated with accurate data from Snowflake.

Snowflake contains the true source of data for the **purchase_date** and **warranty_expiration** fields, which are critical for managing the lifecycle of Windows Server CIs. This integration was chosen over ServiceNow Discovery because Snowflake already holds the most reliable and up-to-date information for these fields. The scheduled job ensures that the CMDB is synchronized daily with Snowflake, providing accurate and consistent data for lifecycle management.

Snowflake-ServiceNow Integration Overview



Step 1: User Creation and Configuration

Create the ServiceNow integration user in Snowflake:

```
CREATE USER SERVICENOW_USER PASSWORD = 'Snow2Service@2025';
```


Create and assign the necessary role:

```
CREATE ROLE SERVICENOW_METADATA_ROLE;
CREATE ROLE SERVICENOW_INTEGRATION_ROLE;
GRANT ROLE SERVICENOW_METADATA_ROLE TO USER SERVICENOW_USER;
GRANT ROLE SERVICENOW_INTEGRATION_ROLE TO USER SERVICENOW_USER;
```

Grant warehouse access to the **SERVICENOW_USER**:

```
GRANT USAGE ON WAREHOUSE COMPUTE_WH TO ROLE SERVICENOW_INTEGRATION_ROLE;
GRANT OPERATE ON WAREHOUSE COMPUTE_WH TO ROLE SERVICENOW_INTEGRATION_ROLE;
```


Grant access to the database and schema:

```
GRANT USAGE ON DATABASE SERVICENOW TO ROLE SERVICENOW_INTEGRATION_ROLE;
GRANT USAGE ON SCHEMA SERVICENOW.PUBLIC TO ROLE SERVICENOW_INTEGRATION_ROLE;
GRANT SELECT ON TABLE SERVICENOW.PUBLIC.CMDB_ENRICHMENT_DATA TO ROLE
SERVICENOW_INTEGRATION_ROLE;
```

Step 2: Security Integration Setup

Create the OAuth security integration using the **ACCOUNTADMIN** role:

```
CREATE SECURITY INTEGRATION MY_SERVICENOW_CLIENT
  TYPE = OAUTH
  ENABLED = TRUE
  OAUTH_CLIENT = CUSTOM
  OAUTH_CLIENT_TYPE = 'CONFIDENTIAL'
  OAUTH_REDIRECT_URI = 'https://dev197363.service-now.com/oauth_redirect.do'
  OAUTH_ISSUE_REFRESH_TOKENS = TRUE
  OAUTH_REFRESH_TOKEN_VALIDITY = 86400
  PRE_AUTHORIZED_ROLES_LIST = ('SERVICENOW_METADATA_ROLE',
'SERVICENOW_INTEGRATION_ROLE');
  BLOCKED_ROLES_LIST = ('ACCOUNTADMIN', 'ORGADMIN', 'SECURITYADMIN');
```


Verify the integration configuration:

```
DESCRIBE SECURITY INTEGRATION MY_SERVICENOW_CLIENT;
```

Retrieve and securely store the client secret:

```
SELECT SYSTEM$SHOW_OAUTH_CLIENT_SECRETS('MY_SERVICENOW_CLIENT');
```

Step 3: REST Message & OAuth 2.0 Setup

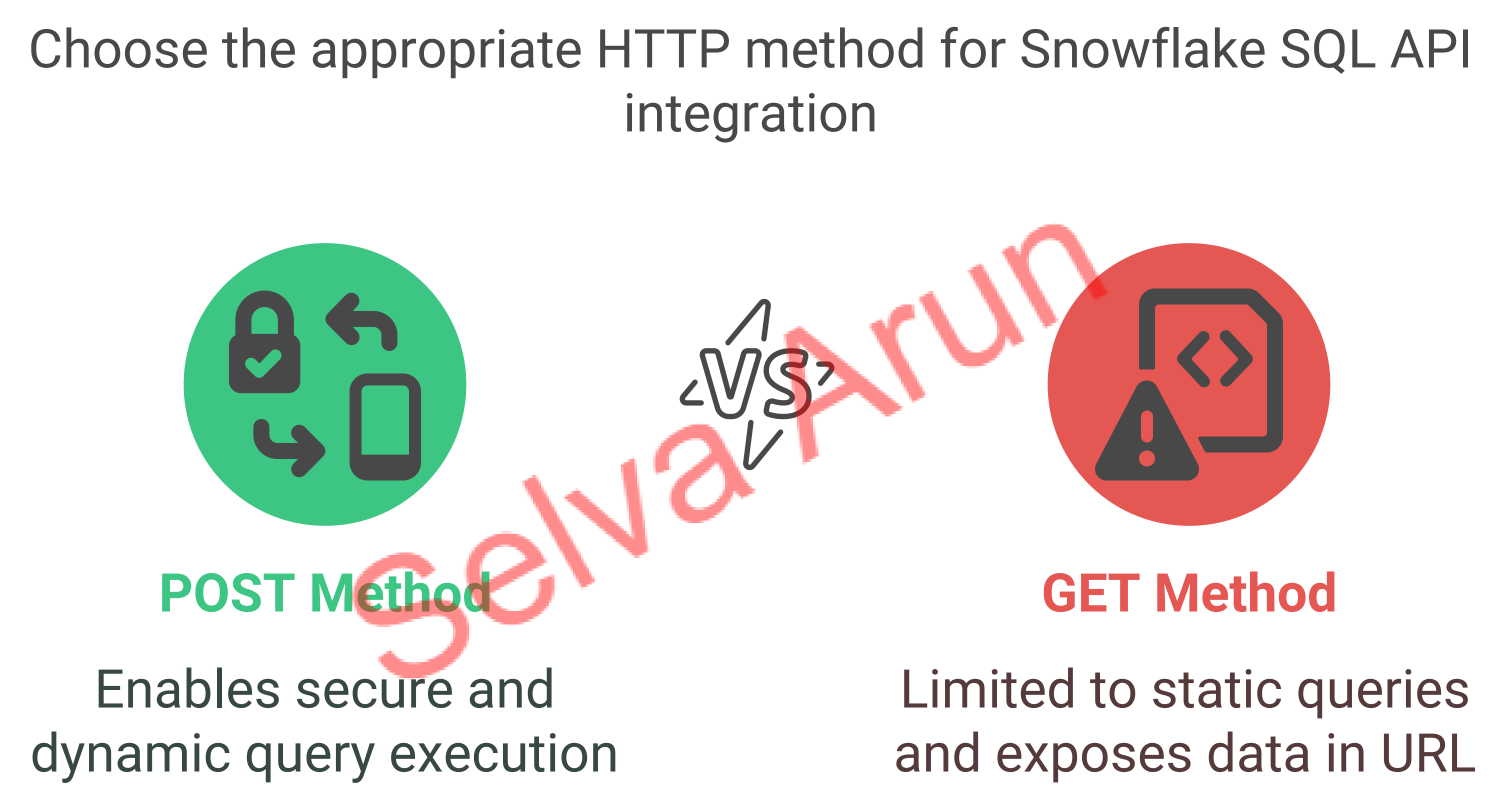
To fetch data from Snowflake, a REST Message was configured in ServiceNow. This REST Message uses the Snowflake SQL API to execute SQL queries and retrieve the required data.

Why Use POST Instead of GET?

The POST method was used for the REST Message because:

1. **SQL Query Execution**: The Snowflake SQL API requires the SQL query to be sent in the request body, which is only possible with the POST method. The GET method does not support sending a request body.
2. **Dynamic Querying**: Using POST allows us to dynamically execute SQL queries, such as selecting specific columns or filtering data based on conditions.
3. **Security**: The POST method ensures that sensitive information, such as SQL queries or parameters, is not exposed in the URL, providing better security compared to GET.

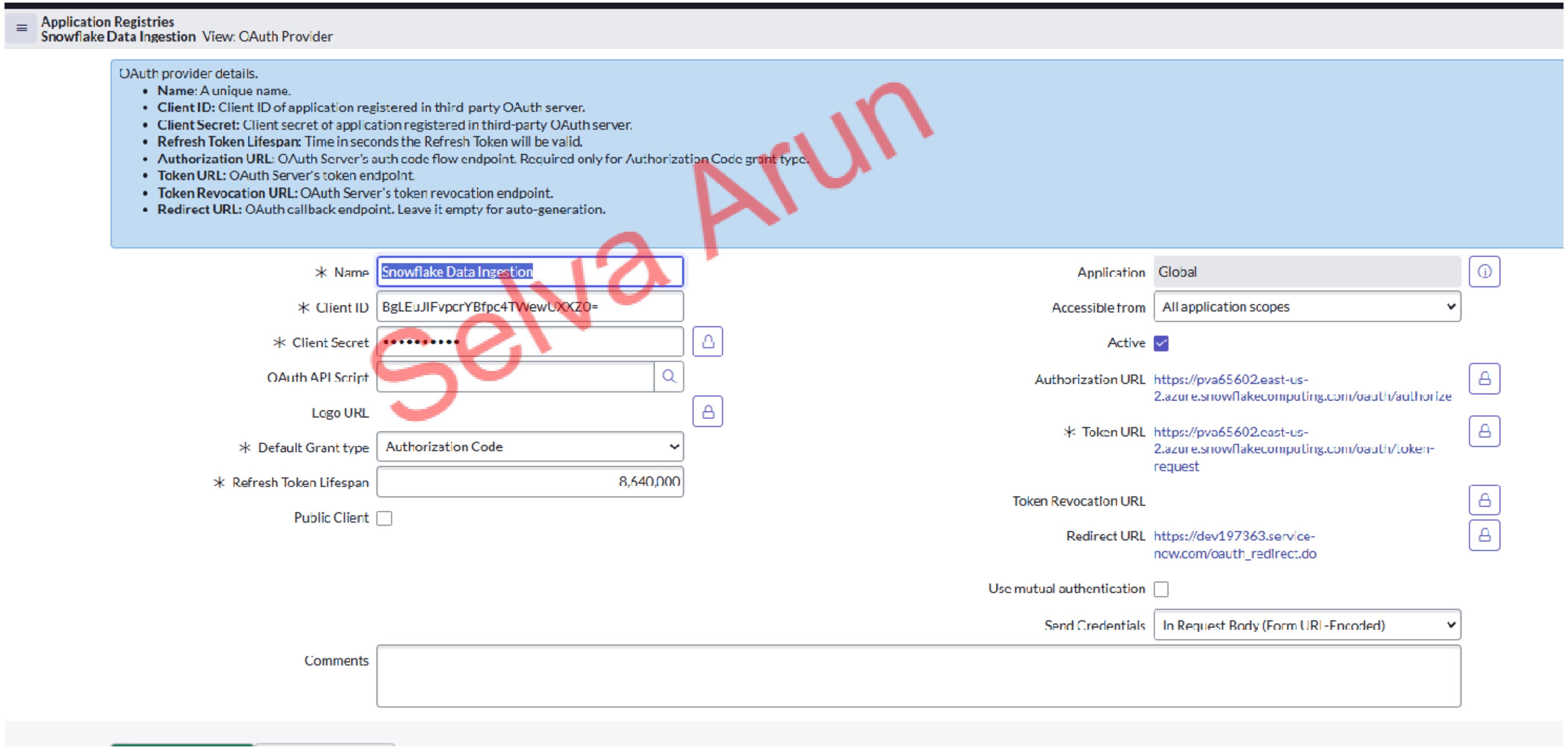
4. **Compliance with Snowflake API**: The Snowflake SQL API is designed to execute queries via POST requests, making it the only valid method for this use case.



OAuth Configuration in ServiceNow

The OAuth configuration was set up in the Application Registry in ServiceNow:

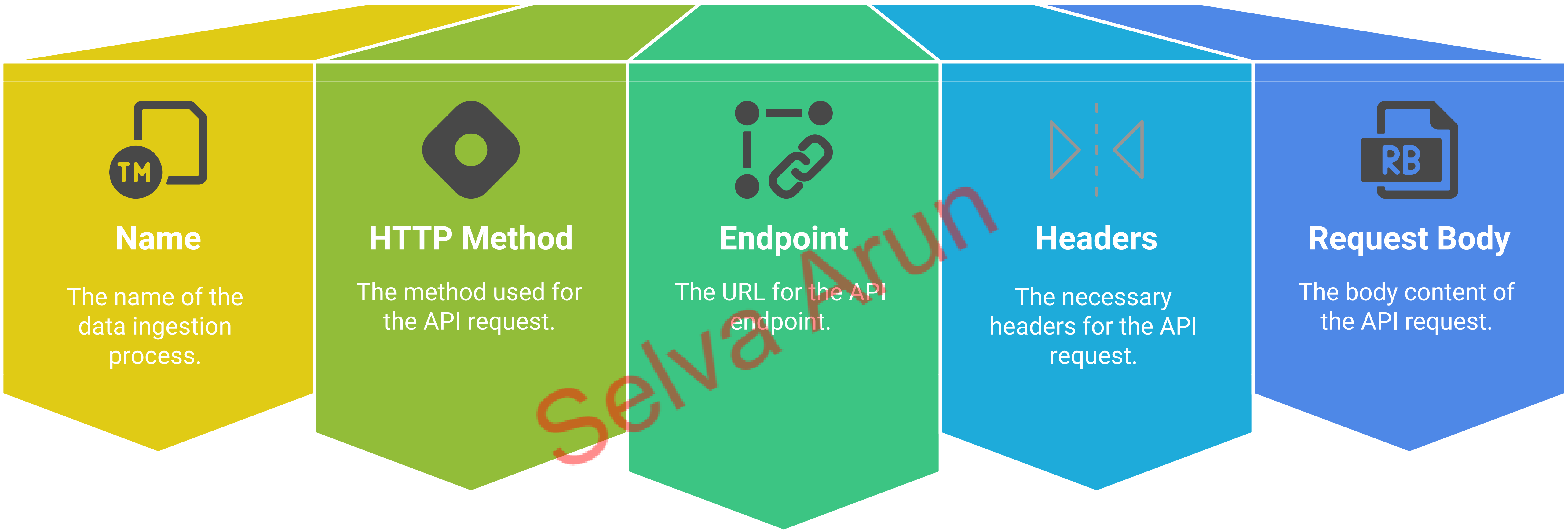
- **Client ID**: BgLEuJIFvpcrYBfpc4TWewUXXZ0=
- **Authorization Endpoint**: https://pva65602.east-us-2.azure.snowflakecomputing.com/oauth/authorize
- **Token Endpoint**: https://pva65602.east-us-2.azure.snowflakecomputing.com/oauth/token-request
- **Redirect URI**: https://dev197363.service-now.com/oauth_redirect.do



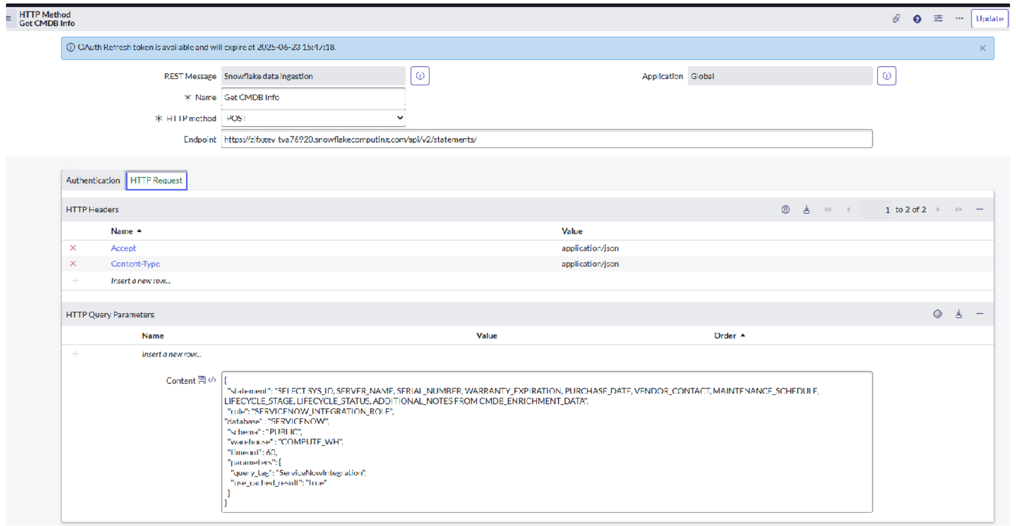
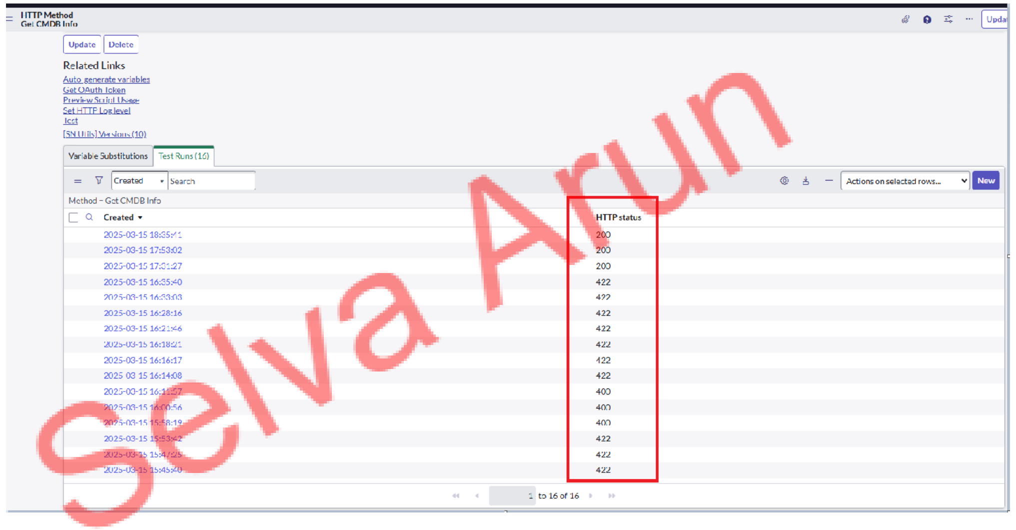
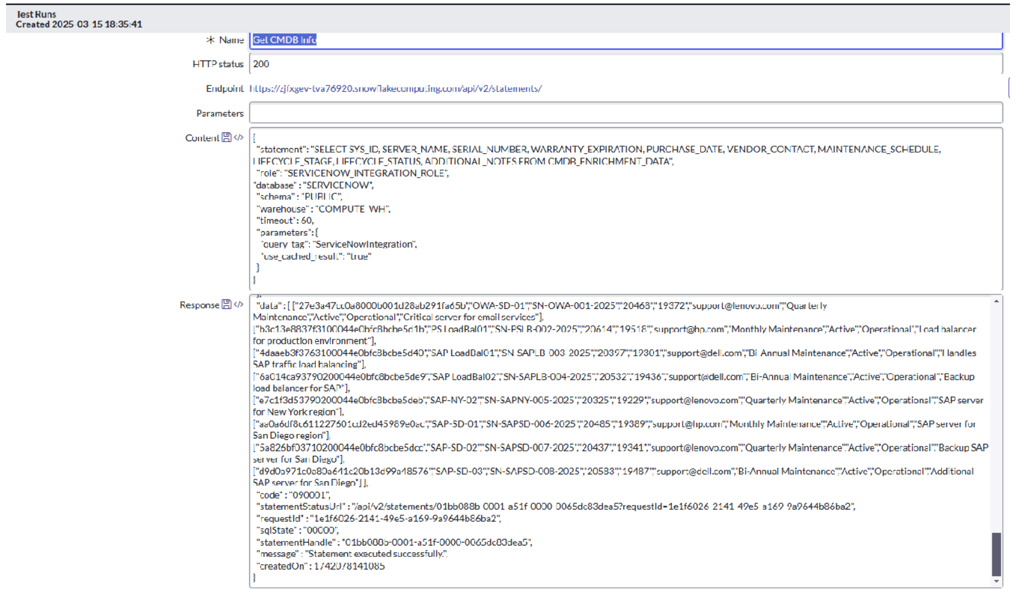
REST Message Configuration

- **Name**: Snowflake data ingestion
- **HTTP Method**: POST
- **Endpoint**: https://zjfxgev-tva76920.snowflakecomputing.com/api/v2/statements/
- **Headers**:
    - Authorization: Bearer token (retrieved via OAuth)
    - Content-Type: application/json
- **Request Body**:

Snowflake Data Ingestion



```
{
  "statement": "SELECT SYS_ID, SERVER_NAME, SERIAL_NUMBER, WARRANTY_EXPIRATION,
PURCHASE_DATE, VENDOR_CONTACT, MAINTENANCE_SCHEDULE, LIFECYCLE_STAGE,
LIFECYCLE_STATUS, ADDITIONAL_NOTES FROM CMDB_ENRICHMENT_DATA",
  "role": "SERVICENOW_INTEGRATION_ROLE",
  "database": "SERVICENOW",
  "schema": "PUBLIC",
  "warehouse": "COMPUTE_WH",
  "timeout": 60,
  "parameters": {
    "query_tag": "ServiceNowIntegration",
    "use_cached_result": "true"
  }
}
```

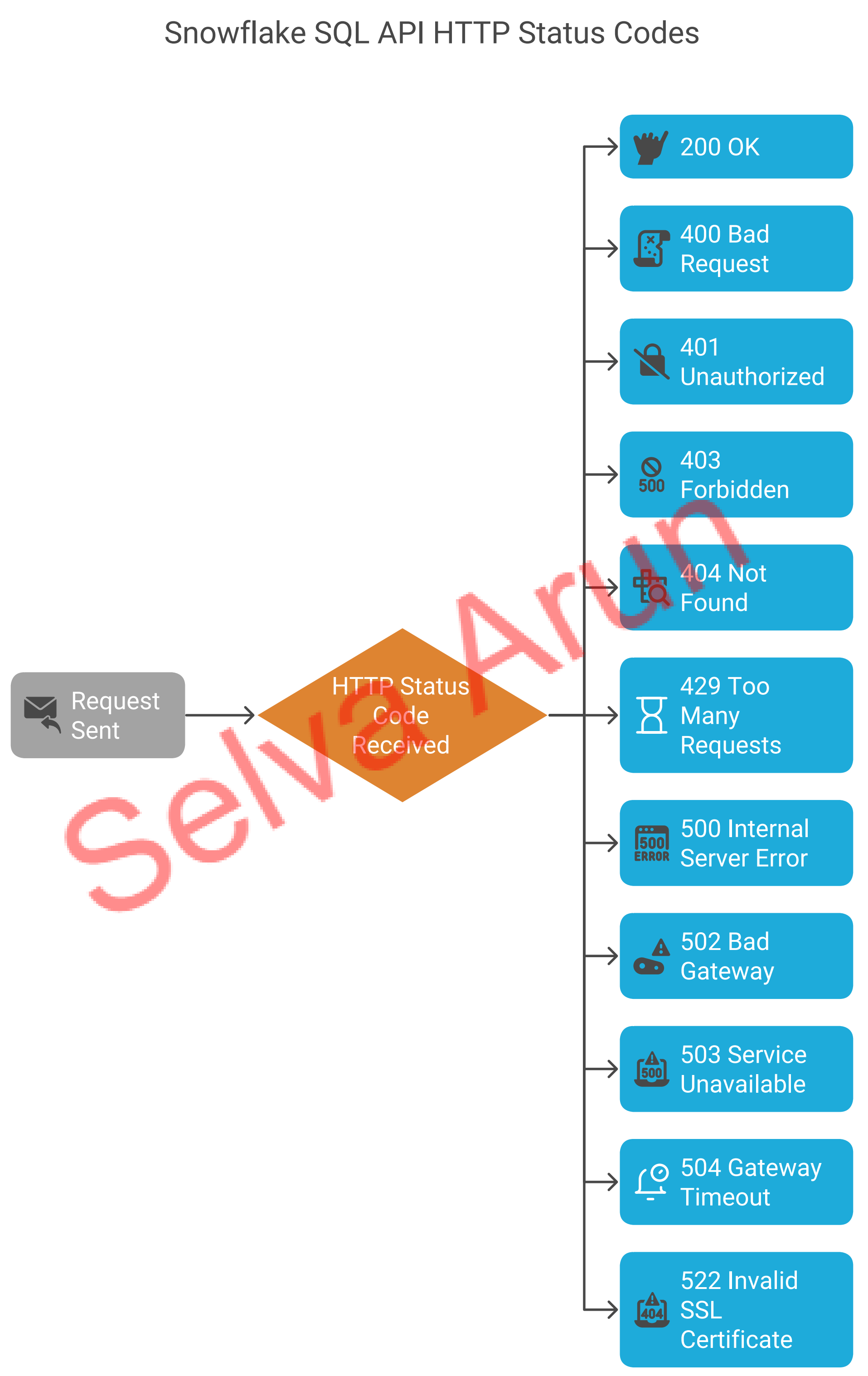

HTTP Method
Get CMDB Info
Update
OAuth Refresh token is available and will expire at 2025-06-23 15:47:18.
REST Message
Snowflake data ingestion
Application
Global
Name
Get CMDB Info
HTTP method
POST
Endpoint
https://zjfxgev-tva76920.snowflakecomputing.com/api/v2/statements/
Authentication
HTTP Request
HTTP Headers
1 to 2 of 2
Name
Value
Accept
application/json
Content-Type
application/json
Insert a new row...
HTTP Query Parameters
Name
Value
Order
insert a new row...
Content
{
"statement": "SELECT SYS_ID, SERVER_NAME, SERIAL_NUMBER, WARRANTY_EXPIRATION, PURCHASE_DATE, VENDOR_CONTACT, MAINTENANCE_SCHEDULE, LIFECYCLE_STAGE, LIFECYCLE_STATUS, ADDITIONAL_NOTES FROM CMDB_ENRICHMENT_DATA",
"role": "SERVICENOW_INTEGRATION_ROLE",
"database" : "SERVICENOW",
"schema" : "PUBLIC",
"warehouse" : "COMPUTE_WH",
"timeout": 60,
"parameters": {
"query_tag": "ServiceNowIntegration",
"use_cached_result": "true"
}
}

HTTP Method
Get CMDB Info
Update
Delete
Related Links
Auto generate variables
Get OAuth Token
Preview Script Usage
Set HTTP Log level
Test
[SN Utils] Versions (10)
Variable Substitutions
Test Runs (16)
Created
Search
Actions on selected rows...
New
Method = Get CMDB Info
Created
HTTP status
2025-03-15 18:35:41
200
2025-03-15 17:53:02
200
2025-03-15 17:31:27
200
2025-03-15 16:35:40
422
2025-03-15 16:33:03
422
2025-03-15 16:28:16
422
2025-03-15 16:21:46
422
2025-03-15 16:18:21
422
2025-03-15 16:16:17
422
2025-03-15 16:14:08
422
2025-03-15 16:11:57
400
2025-03-15 16:00:56
400
2025-03-15 15:58:19
400
2025-03-15 15:53:42
422
2025-03-15 15:47:25
422
2025-03-15 15:45:40
422
1 to 16 of 16
Selva Arun

Test Runs
Created 2025 03 15 18:35:41
Name
Get CMDB Info
HTTP status
200
Endpoint
https://zjfxgev-tva76920.snowflakecomputing.com/api/v2/statements/
Parameters
Content
{
"statement": "SELECT SYS_ID, SERVER_NAME, SERIAL_NUMBER, WARRANTY_EXPIRATION, PURCHASE_DATE, VENDOR_CONTACT, MAINTENANCE_SCHEDULE, LIFECYCLE_STAGE, LIFECYCLE_STATUS, ADDITIONAL_NOTES FROM CMDB_ENRICHMENT_DATA",
"role": "SERVICENOW_INTEGRATION_ROLE",
"database" : "SERVICENOW",
"schema" : "PUBLIC",
"warehouse" : "COMPUTE_WH",
"timeout": 60,
"parameters": {
"query_tag": "ServiceNowIntegration",
"use_cached_result": "true"
}
}
Response
"data" : [["27e3a47cc0a8000b001d28ab291fa65b","OWA-SD-01","SN-OWA-001-2025","20468","19372","support@lenovo.com","Quarterly Maintenance","Active","Operational","Critical server for email services"],
["b3c13e8837f3100044e0bfc8bcbe5d1b","PS LoadBal01","SN-PSLB-002-2025","20614","19518","support@hp.com","Monthly Maintenance","Active","Operational","Load balancer for production environment"],
["4daaeb3f3763100044e0bfc8bcbe5d40","SAP LoadBal01","SN SAPLB 003 2025","20397","19301","support@dell.com","Bi Annual Maintenance","Active","Operational","Handles SAP traffic load balancing"],
["6a014ca93790200044e0bfc8bcbe5de9","SAP LoadBal02","SN-SAPLB-004-2025","20532","19436","support@dell.com","Bi-Annual Maintenance","Active","Operational","Backup load balancer for SAP"],
["e7c1f3d53790200044e0bfc8bcbe5deb","SAP-NY-02","SN-SAPNY-005-2025","20325","19229","support@lenovo.com","Quarterly Maintenance","Active","Operational","SAP server for New York region"],
["aa0a6df8c611227601cd2ed45989e0ac","SAP-SD-01","SN-SAPSD-006-2025","20485","19389","support@hp.com","Monthly Maintenance","Active","Operational","SAP server for San Diego region"],
["5a826bf03710200044e0bfc8bcbe5dcc","SAP-SD-02","SN-SAPSD-007-2025","20437","19341","support@lenovo.com","Quarterly Maintenance","Active","Operational","Backup SAP server for San Diego"],
["d9d0a971c0a80a641c20b13d99a48576","SAP-SD-03","SN-SAPSD-008-2025","20583","19487","support@dell.com","Bi-Annual Maintenance","Active","Operational","Additional SAP server for San Diego"]],
"code" : "090001",
"statementStatusUrl" : "/api/v2/statements/01bb088b 0001 a51f 0000 0065dc83dea5?requestId=1e1f6026 2141 49e5 a169 9a9644b86ba2",
"requestId" : "1e1f6026-2141-49e5-a169-9a9644b86ba2",
"sqlState" : "00000",
"statementHandle" : "01bb088b-0001-a51f-0000-0065dc83dea5",
"message" : "Statement executed successfully.",
"createdOn" : 1742078141085
}

Snowflake API Documentation

For more details about the Snowflake SQL API, refer to the official documentation: [Snowflake SQL API Documentation](https://docs.snowflake.com/en/developer-guide/sql-api/index.html).

HTTP Status Codes and Error Messages

When interacting with the Snowflake SQL API, you may encounter various HTTP status codes. Below are some common status codes and their meanings:


Snowflake SQL API HTTP Status Codes
Request Sent
HTTP Status Code Received
200 OK
400 Bad Request
401 Unauthorized
403 Forbidden
500
404 Not Found
429 Too Many Requests
500 Internal Server Error
500 ERROR
502 Bad Gateway
503 Service Unavailable
500
504 Gateway Timeout
522 Invalid SSL Certificate
404

Step 4: Scheduled Job Implementation

To automate the update of Windows Server CI records in the CMDB, a scheduled job was created in ServiceNow. This job fetches data from Snowflake, converts the date fields from Epoch format to standard YYYY-MM-DD format, and updates the **warranty_expiration** and **purchase_date** fields in the **cmdb_ci_win_server** table.

Why Use a Scheduled Job?

The scheduled job was implemented to:

- Automate the data synchronization process between Snowflake and ServiceNow.
- Explore the use case of integrating Snowflake data with the CMDB for better lifecycle management of Windows Server CIs.
- Ensure the CMDB is updated daily with the latest data from Snowflake.

The job runs at 12:00 AM daily to ensure the CMDB is always up-to-date.

Script for Scheduled Job

The following script was used in the scheduled job to fetch data from Snowflake, convert the dates, and update the CMDB:

```
try {
    var r = new sn_ws.RESTMessageV2('Snowflake data ingestion', 'Get CMDB
Info');
    var response = r.execute();
    var responseBody = response.getBody();
    var httpStatus = response.getStatusCode();
    if (httpStatus == 200) {
        var responseData = JSON.parse(responseBody);
        var data = responseData.data;
        for (var i = 0; i < data.length; i++) {
            var row = data[i];
            var serialNumber = row[2];
            var warrantyExpiration = row[3];
            var purchaseDate = row[4];
            if (!serialNumber) continue;
            var warrantyDateStr = convertDaysToDate(warrantyExpiration);
            var purchaseDateStr = convertDaysToDate(purchaseDate);
            var gr = new GlideRecord('cmdb_ci_win_server');
            gr.addQuery('serial_number', serialNumber);
            gr.query();
            while (gr.next()) {
                if (warrantyDateStr) {
                    var warrantyDate = new GlideDateTime();
                    warrantyDate.setValue(warrantyDateStr);
                    gr.warranty_expiration = warrantyDate;
                }
                if (purchaseDateStr) {
                    var purchaseDateTime = new GlideDateTime();
                    purchaseDateTime.setValue(purchaseDateStr);
                    gr.purchase_date = purchaseDateTime;
                }
                gr.update();
            }
        }
    }
} catch (ex) {
    gs.print('Error occurred: ' + ex.message);
}

function convertDaysToDate(daysSinceEpoch) {
    if (!daysSinceEpoch || isNaN(daysSinceEpoch)) return null;
    var epoch = new Date(1970, 0, 1);
    epoch.setDate(epoch.getDate() + parseInt(daysSinceEpoch));
    return epoch.toISOString().split('T')[0];
}
```

Why Dates Are in Epoch Format

The date fields (**purchase_date** and **warranty_expiration**) in Snowflake are stored in **Epoch format** (also known as Unix time). Epoch format represents the number of days or seconds that have elapsed since **January 1, 1970 (UTC)**. This format is commonly used in databases and APIs because it is compact, easy to store, and avoids issues with time zones.

In this integration, Snowflake returns the date fields in **days since Epoch** format. This is because Snowflake's internal date representation often uses days for simplicity and efficiency when working with date calculations.

How Dates Were Converted

To make the date fields usable in ServiceNow, they were converted from **days since Epoch** to the standard **YYYY-MM-DD** format. This conversion was necessary because ServiceNow requires date fields to be in a human-readable format for proper storage and display in the CMDB.

The following JavaScript function was used in the scheduled job to perform the conversion:

```
function convertDaysToDate(daysSinceEpoch) {
    if (!daysSinceEpoch || isNaN(daysSinceEpoch)) {
        return null; // Return null if the input is invalid
    }
    var epoch = new Date(1970, 0, 1);
    epoch.setDate(epoch.getDate() + parseInt(daysSinceEpoch));
    return epoch.toISOString().split('T')[0];
}
```

Testing:

Windows Servers | Name | Search | Actions on selected rows... | New

Name ▲	Serial number	Warranty expiration	Purchased	Vendor	Maintenance schedule	Life Cycle Stage	Life Cycle Stage Status
OWA-SD-01	SN-OWA-001-2025	2026-01-15	2023-01-15	Lenovo	(empty)	(empty)	(empty)
PS LoadBal01	SN-PSLB-002-2025	2026-06-10	2023-06-10	Lenovo	(empty)	(empty)	(empty)
SAP LoadBal01	SN-SAPLB-003-2025	2025-11-05	2022-11-05	Lenovo	(empty)	(empty)	(empty)
SAP LoadBal02	SN-SAPLB-004-2025	2026-03-20	2023-03-20	Lenovo	(empty)	(empty)	(empty)
SAP-NY-02	SN-SAPNY-005-2025	2025-08-25	2022-08-25	Lenovo	(empty)	(empty)	(empty)
SAP-SD-01	SN-SAPSD-006-2025	2026-02-01	2023-02-01	Lenovo	(empty)	(empty)	(empty)
SAP-SD-02	SN-SAPSD-007-2025	2025-12-15	2022-12-15	Lenovo	(empty)	(empty)	(empty)
SAP-SD-03	SN-SAPSD-008-2025	2026-05-10	2023-05-10	Lenovo	(empty)	(empty)	(empty)

```sql
SELECT SYS_ID, SERVER_NAME, SERIAL_NUMBER, WARRANTY_EXPIRATION, PURCHASE_DATE, VENDOR_CONTACT, MAINTENANCE_SCHEDULE, LIFECYCLE_STAGE, LIFECY
FROM CMDB_ENRICHMENT_DATA
```

Results | Chart

	SYS_ID	SERVER	SERIAL_NUMB	WARRANTY_EXP	PURCHAS	VENDOR_CO	MAINTENANCE_SC	LIFECY
1	27e3a47cc0a8000b001d28ab291fa65	OWA-SD-01	SN-OWA-001-2025	2026-01-15	2023-01-15	support@lenovo.com	Quarterly Maintenance	Active
2	b3c13e8837f3100044e0bfc8bcbe5d1	PS LoadBal01	SN-PSLB-002-2025	2026-06-10	2023-06-10	support@hp.com	Monthly Maintenance	Active
3	4daaeb3f3763100044e0bfc8bcbe5d4	SAP LoadBal01	SN-SAPLB-003-2025	2025-11-05	2022-11-05	support@dell.com	Bi-Annual Maintenance	Active
4	6a014ca93790200044e0bfc8bcbe5de	SAP LoadBal02	SN-SAPLB-004-2025	2026-03-20	2023-03-20	support@dell.com	Bi-Annual Maintenance	Active
5	e7c1f3d53790200044e0bfc8bcbe5de	SAP-NY-02	SN-SAPNY-005-2025	2025-08-25	2022-08-25	support@lenovo.com	Quarterly Maintenance	Active
6	aa0a6df8c611227601cd2ed45989e0a	SAP-SD-01	SN-SAPSD-006-2025	2026-02-01	2023-02-01	support@hp.com	Monthly Maintenance	Active
7	5a826bf03710200044e0bfc8bcbe5dc	SAP-SD-02	SN-SAPSD-007-2025	2025-12-15	2022-12-15	support@lenovo.com	Quarterly Maintenance	Active
8	d9d0a971c0a80a641c20b13d99a4857	SAP-SD-03	SN-SAPSD-008-2025	2026-05-10	2023-05-10	support@dell.com	Bi-Annual Maintenance	Active

Conclusion

This integration ensures that the CMDB is updated daily with accurate data from Snowflake. The conversion of dates from Epoch format to **YYYY-MM-DD** ensures compatibility with ServiceNow's data requirements, providing a seamless and reliable solution for managing Windows Server lifecycle information.

Data Integration Transformation